

Data Storytelling & Statistical Validation

Online Retail Analysis

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Objective

- Analyze retail sales data
- Identify key revenue drivers
- Understand customer behavior
- Provide business recommendations

Dataset Overview

- **Online Retail Dataset**

- **Key Features:**

Country

CustomerID

Quantity

UnitPrice

InvoiceDate

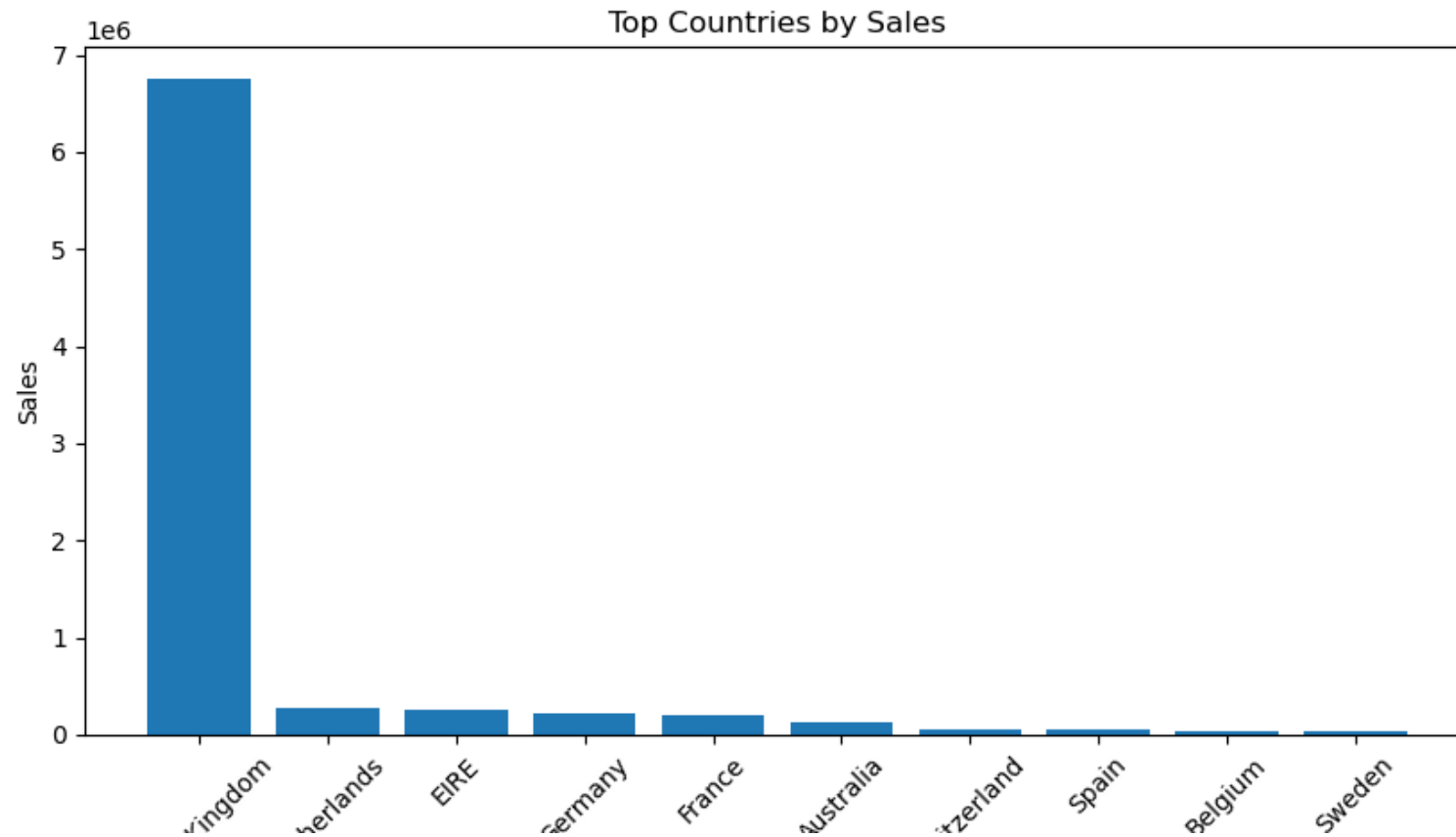
- **Created Feature:**

$\text{Sales} = \text{Quantity} \times \text{UnitPrice}$

Analysis Performed

- Country-wise sales analysis
- Top product analysis
- Monthly sales trends
- Customer contribution analysis

Top Countries by Sales



United Kingdom dominates overall sales performance

Key Insights

- United Kingdom generates highest sales
- Few products dominate total revenue
- Sales show monthly fluctuations
- Small group of customers drives majority revenue

Business Problems

- High dependency on **limited countries**
- Revenue concentrated among **few customers**
- **Uneven sales distribution**

Hypothesis

High-value customers spend more than low-value customers

Statistical Test

- Test Used: Independent T-Test
- Significance Level: 0.05
- P-Value: $8.73e-38$

Result

- Reject Null Hypothesis
- High-value customers spend significantly more than low-value customers

Recommendations

- Focus on high-value customers
- Expand business to new countries
- Promote top-selling products
- Apply seasonal marketing strategies

Conclusion

- Data-driven insights improve decision making
- Statistical validation strengthens analysis
- Business can increase revenue using insights